5. Section #5 Coding Questions

1 2

Previous Section Next

Attempted: 2/2

Question 1

Revisit Later

How to Attempt?

Write a function to round all decimals in a list down to their nearest integer i.e. the output should be a list of integers. For example, if input1 is [1.2, 4.6, 3.5] the function should return [1,4,3].

PYTHON2 Compiler Python 2.7

```
# Read only region start
class UserMainCode(object):
    @classmethod
    def RoundDown(cls, input1):
        '''
        input1 : List[Float]

        Expected return type : List[Integer]
        '''
        # Read only region end
        # Write code here
        n=len(input1)
        for i in range(n):
            input1[i]=int(input1[i])
        return input1
```

IntelliSense has loaded

Use Custom Input

Compile and Test Submit Code

Question 2

Revisit Later

Which library is commonly used for creating plots in Python?

Select an option

Clear Response

- ○ matplotlib
- ○ seaborn
- ○ plotly
- ◉ All of the above

Question 3

Revisit Later

Which attribute of a Pandas DataFrame returns its column labels?

Select an option

Clear Response

- (●) df.columns
- ( ) df.colnames
- ( ) df.labels
- ( ) df.column_names

Question 3

Revisit Later

You need to find the rank of employees within each department based on their salary, but you want to account for ties (i.e., employees with the same salary should have the same rank). Which SQL query will achieve this?

Select an option

Clear Response

- SELECT EmployeeID, Department, Salary,
  RANK() OVER (PARTITION BY Department ORDER BY Salary DESC) AS Rank
  FROM Employees
- SELECT EmployeeID, Department, Salary,
  DENSE_RANK() OVER (PARTITION BY Department ORDER BY Salary DESC) AS Rank
  FROM Employees
- SELECT EmployeeID, Department, Salary,
  ROW_NUMBER() OVER (PARTITION BY Department ORDER BY Salary DESC) AS Rank
  FROM Employees
- SELECT EmployeeID, Department, Salary,
  RANK() OVER (ORDER BY Salary DESC) AS Rank
  FROM Employees

Question 5

Revisit Later

How do you group data by 'site_id' and calculate the mean of 'consumption' in a DataFrame called df?

Select an option

Clear Response

- df.groupby('site_id')['consumption'].mean()
- df.groupby('site_id').mean('consumption')
- df.groupby('site_id').agg({'consumption': 'mean'})
- (●) All of the above

Question 2

Revisit Later

Your daily data import suddenly shows exactly 50% fewer records than usual. The MOST likely cause is:

Select an option

Clear Response

- ○ Random sampling error
- ◉ One of two data sources failed to load
- ○ Database performance issues
- ○ Data corruption

Question 7

Revisit Later

Consider the following:

```
df = pd.DataFrame({
    'A': [1, 2, 3, 4],
    'B': [5, 6, 7, 8]
})
result = df.apply(lambda x: x**2 if x.name == 'A' else x+1)
```

What will be the output in the result dataframe?

Select an option

Clear Response

- ○ Squares of both column 'A' and 'B' values
- ○ Original DataFrame unchanged
- ○ All values incremented by 1
- ◉ Squares of column 'A' values and column 'B' values incremented by 1

Question 1

Revisit Later

For a table with millions of energy records for consumption sites queried frequently by date ranges, what's the best indexing strategy?

Select an option

Clear Response

- ○ Create index on reading_date
- ◉ Create composite index on (site_id, reading_date)
- ○ Partition table by date
- ○ All of the above

Question 4

Revisit Later

You have an SQL table called Sales, with the columns Date, Region, SalesAmount. Which of the following SQL queries will return the sum of SalesAmount for each Region, along with the cumulative sum over all the rows for each region, ordered by Date?

Select an option

Clear Response

- (●) SELECT Region, SalesAmount, SUM(SalesAmount) OVER (PARTITION BY Region ORDER BY Date) AS CumulativeSales FROM Sales
- ( ) SELECT Region, SalesAmount, SUM(SalesAmount) OVER (ORDER BY Region, Date) AS CumulativeSales FROM Sales
- ( ) SELECT Region, SalesAmount, SUM(SalesAmount) OVER (PARTITION BY Region) AS CumulativeSales FROM Sales
- ( ) SELECT Region, SalesAmount, SUM(SalesAmount) OVER (ORDER BY Date) AS CumulativeSales FROM Sales

Question 3

Revisit Later

You want an LLM to analyze data and THEN explain its reasoning. Due to how autoregressive models work, you should:

Select an option

Clear Response

- (●) Ask for reasoning first, then the analysis
- ( ) Ask for analysis first, then reasoning
- ( ) The order doesn't matter
- ( ) Let the LLM decide since the order depends on the task

5. Section #5 Coding Questions

1 2

Previous Next Section

Attempted: 2/2

Question 2

Revisit Later

How to Attempt?

You are given two integers a and b in a list [a,b] . In one move, you can add or subtract any integer from 1 to 10 to a. Write a function to determine the minimum number of moves required to make a equal to b. For example, if input1 is [5, 27], the function should return 3.

PYTHON3 Compiler: Python 3.11

```
# Read only region start
class UserMainCode(object):
    @classmethod
    def MinMoves(cls, input1):
        '''
        input1 : List[Integer]

        Expected return type : int
        '''
        # Read only region end
        # Write code here
        a,b=input1
        diff=abs(a-b)
        cnt=diff//10
        if diff%10 !=0:
            return cnt+1
        return cnt
```

Use Custom Input

Compile and Test Submit Code

Question 2

Revisit Later

Which data type should you use to store offshore wind project names efficiently?

Select an option

Clear Response

- ○ CHARMAX()
- ○ VARCHAR(N) where N is finite real number
- ◉ VARCHAR(N) where N is finite natural number
- ○ VARCHAR(MAXIMUM)

Question 2

Revisit Later

What function in Pandas is used to drop rows or columns with missing values?

Select an option

Clear Response

- (●) dropna()
- ( ) fillna()
- ( ) isnull()
- ( ) notnull()

Question 6

Revisit Later

Suppose you have an employees table with columns employee_id, name, and manager_id.
Which SQL query will list the name of each employee along with their manager's name?

Select an option

Clear Response

- (●) SELECT e.name AS employee, m.name AS manager
  FROM employees e
  LEFT JOIN employees m ON e.manager_id = m.employee_id;
- ( ) SELECT name, manager_id
  FROM employees;
- ( ) SELECT e.name, e.manager_id
  FROM employees e
  JOIN employees m ON e.employee_id = m.manager_id;
- ( ) SELECT name AS employee, name AS manager
  FROM employees;

Question 4 Revisit Later

How do you create a decorator that retries failed API calls for gas pipeline data with exponential backoff?

Select an option Clear Response

- ○ Retry immediately on failure
- ◉ Wait 2^(attempt_number)seconds between retries
- ○ Use random retry delays
- ○ Retry only once

Question 1 Revisit Later

What is the primary goal of the softmax function in a neural network classifier?

Select an option Clear Response

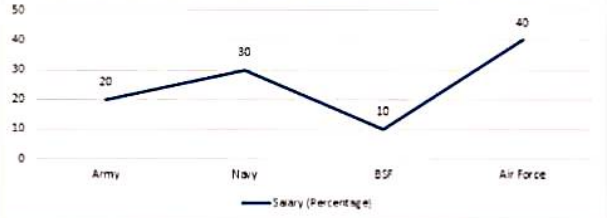
- ○ To introduce non-linearity
- ○ To reduce dimensionality
- ◉ To convert scores into probabilities
- ○ To perform feature scaling



The number of persons who joined Indian defense service in April, May, June, July, August, September and October were 1800, 1500, 1600, 2000, 2200, 2400 and 2500 respectively.

NOTE: Each person had joined service on the 1 st date of the concerned month.

The line graph given below shows the percentage of which the salary of each employee of the other 4 divisions were more than that of CRPF whose salary per employee was ₹ 10,000 per month.



NOTE: Each employee gets salary at the end of the month. For example, the salary for April was given at the end of April, i.e. 30 th April.

The number of persons who joined BSF was approximately was what percentage of the total persons who had joined defense services during the given period?

Select an option

Clear Response

- 29.28%
- 23.57%
- 32.57%
- 20.26%

Question 8

Revisit Later

When would Selenium be genuinely necessary over simpler methods?

Select an option

Clear Response

- ○ When you need to handle cookies
- ◉ When you need to interact with elements (clicking, scrolling)
- ○ When the site requires headers
- ○ When scraping multiple pages

Question 4

Revisit Later

How do you select a column named 'consumption per_day' from a DataFrame df?

Select an option

Clear Response

- A. df['consumption per_day']
- B. df.consumption per_day
- Both A and B
- C. None of the above

Question 5

Revisit Later

You need to validate gas composition data where each reading must sum to 100%. Which validation approach is most efficient?

Select an option

Clear Response

- Load all data into memory, then filter
- Use generator with yield for valid readings only
- Create pandas DataFrame and filter
- Convert to numpy array and validate

Question 8

Revisit Later

How do you detect duplicate rows in a DataFrame df based on the 'timestamp' column?

Select an option

Clear Response

- df.duplicated('timestamp')
- df.duplicated(subset='timestamp') (selected)
- df.duplicated(['timestamp'])
- All of the above

Question 3 Revisit Later

You have a 20GB CSV of wind turbine data. Which approach processes it without memory errors?

Select an option Clear Response

- ○ Load entire file into memory with pd.read_csv()
- ◉ Process in chunks using chunksize parameter
- ○ Read file line by line into a list
- ○ Use multiprocessing to split the file

Question 1

Revisit Later

How do you sort a DataFrame df by the 'date' column in ascending order?

Select an option

Clear Response

- (●) df.sort_values('date')
- ( ) df.sort('date')
- ( ) df.order_by('date')
- ( ) df.sort_index('date')

Question 6

Revisit Later

You have a DataFrame with 10 million rows. You need to filter for rows where column 'status' equals 'active'. Which approach is fundamentally most efficient?

Select an option

Clear Response

- df[df['status'] == 'active']
- df.query("status == 'active'")
- df.loc[df['status'] == 'active']
- They're all essentially equivalent for this operation

Question 7

Revisit Later

How do you create a line plot of 'consumption' over time using matplotlib?

Select an option

Clear Response

- (●) plt.plot(df['date'], df['consumption'])
- ( ) plt.scatter(df['date'], df['consumption'])
- ( ) plt.bar(df['date'], df['consumption'])
- ( ) plt.hist(df['consumption'])

Question 6

Revisit Later

How do you handle API pagination for renewable energy project data when you don't know the total number of pages?

Select an option

Clear Response

- ○ Guess a large number of pages
- ◉ Loop until empty response is received
- ○ Request all pages at once
- ○ Use a fixed range of 1-1000

Question 5

Revisit Later

Given a dataset where RANK() is applied over the Score column in descending order, how does RANK() handle tied scores?

Select an option

Clear Response

- (●) It assigns the same rank to all tied values but does not skip subsequent ranks.
- ( ) It assigns a unique rank to each row regardless of ties.
- ( ) It assigns the same rank to all tied values and skips subsequent ranks.
- ( ) It randomly assigns ranks to tied values.

Question 8

Revisit Later

When would Selenium be genuinely necessary over simpler methods?

Select an option

Clear Response

- ○ When you need to handle cookies
- ◉ When you need to interact with elements (clicking, scrolling)
- ○ When the site requires headers
- ○ When scraping multiple pages

Question 2

Revisit Later

Six friends, Mathew, Kate, Maria, Bruno, Jose, and June, ordered a medium size pizza which had six slices. As per their order and choices of toppings, the delivered pizza had different toppings on different slices. Once the pizza was delivered, they sat around a circular table along their respective slices. Mathew had ordered for pineapple topping. The person sitting to the left of June had ordered for pepper topping. The person who added Jalapeno sat to the immediate right of the one who had ordered for spinach topping. Maria sat opposite to the person who had ordered for pepper topping. Jose and Bruno had ordered for onion and pepper toppings (not necessarily in the same order). One topping was broccoli. June and Maria had not ordered for spinach toppings. Who sat to the left of Maria?

Select an option

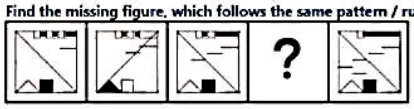
Clear Response

- ○ Bruno
- ◉ Jose
- ○ Mathew
- ○ Kate

Question 1

Revisit Later

Find the missing figure, which follows the same pattern / rule and mark it as answer.

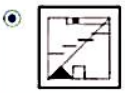


Select an option

Clear Response







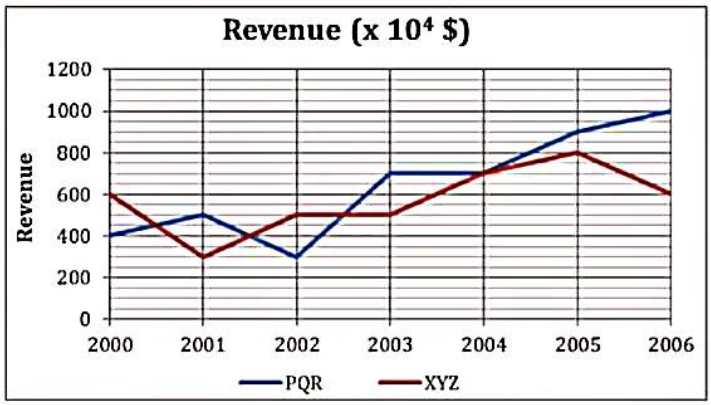




Question 2

Revisit Later

The graph given below shows the revenue earned by two companies 'PQR'and 'XYZ' over the years.



For how many of the given years, the company PQR witnessed a growth of 25% or higher in the revenue generated over the previous year?

Select an option

Clear Response

- (●) 4
- ( ) 3
- ( ) 2
- ( ) 1

Question 1

Revisit Later

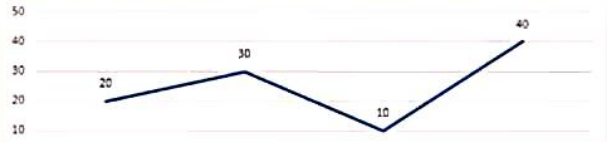
Directions: The bar graph below shows the percentage breakup of persons who joined Indian Defence Service among five divisions – CRPF, BSF, Army, Navy and Air Force during the period April' 2016 – October' 2016.



The number of persons who joined Indian defense service in April, May, June, July, August, September and October were 1800, 1500, 1600, 2000, 2200, 2400 and 2500 respectively.

NOTE: Each person had joined service on the 1 ST date of the concerned month.

The line graph given below shows the percentage of which the salary of each employee of the other 4 divisions were more than that of CRPF whose salary per employee was ₹ 10,000 per month.



Select an option

Clear Response

- ○ 29.28%
- ○ 23.57%
- ○ 32.57%
- ◉ 20.26%

Question 3

Revisit Later

A college has 300 students and all the students use at least one email service from among Yahoo Mail, Gmail and Rediffmail.
A total of 62 students use just one email service, whereas students using all the three are 2 more than this number. The total number of students using Yahoo Mail is 51 more than the total number of students using Gmail. The number of students using Gmail only and Gmail and Rediffmail both, but not Yahoo Mail is 20 and 27 respectively.
Students using Yahoo Mail and Rediffmail both, but not Gmail are 12.5% less than those using all the 3 services.
The total number of students using Rediffmail is 147.

How many students are using both Yahoo Mail and Gmail but not Rediffmail?

Select an option

Clear Response

- (●) 42
- ( ) 91
- ( ) 11
- ( ) Cannot be determined